MissFlow: Flow Matching for Uncertainty-Quantified Imputation of Missing Tabular Data

Joonyoung Jeong Sunil Hwang Younghyun Kwon
Korea Military Academy

ICML International Conference On Machine Learning
The 2nd Workshop on Epistemic Intelligence in Machine Learning

Motivation

❑ The Challenge of Missing Data

✓ Missing tabular data is a pervasive issue (e.g., survey non-response, sensor dropout) that **distorts statistical estimates and invalidates inferences** if left unaddressed.

❑ The Flaw of Single Imputation

✓ Replacing missing entries with a single point estimate systematically **underestimates variance** and leads to **over-confident conclusions**.

❑ The Multiple Imputation (MI) Bottleneck

✓ The principled solution is Multiple Imputation (MI), which draws several completions to propagate uncertainty. However, MI is only practical if sampling completions is **computationally cheap.**

✓ Uncertainty quantification (UQ) via multiple imputation (MI) becomes **computationally prohibitive when using diffusion models**, as generating dozens of required completions demands 50 to 150 NFEs each.

❑ The MissFlow Solution

✓ Because flow matching models transport noise to data along near-straight trajectories for fast, accurate integration, **MissFlow completes records in only ≈ 20 Euler steps,** rendering multiple imputation cheap and uncertainty quantification routine.

✓ To fix the under-coverage commonly found in raw intervals, MissFlow applies a **post-hoc conformal calibration layer that guarantees reliable, finite-sample coverage** without slowing down the fast ODE sampler.

Methodology

❑ Masked Conditional Flow-Matching (CFM)

✓ MissFlow leverages continuous-time normalizing flows to map simple noise distributions to complex tabular data distributions, explicitly conditioning on the observed features.

✓ **Computational Efficiency:** Requires only ≈ 20 Euler steps for completion. This drastic reduction in NFEs compared to standard diffusion makes Multiple Imputation (MI) computationally cheap.

❑ Post-Hoc Conformal Calibration

✓ For each feature j, we compute a nonconformity score over the n_j calibration rows where the feature is originally missing (but for which we know the true held-out value x_j):

$$r^{(j)} = \frac{|x_j - \hat{c}_j|}{\hat{h}_j},$$

where $\hat{c}_j$ and $\hat{h}_j$ are the point estimate, and raw half-width of the uncalibrated interval produced by the generative draws, respectively.

✓ **Calibration Factor (λ_j):** We then calculate a feature-specific scaling factor, λ_j, by taking the empirical quantile of the nonconformity scores. Specifically, λ_j is set to the $\lceil (n_j+1)(1-\alpha) \rceil$-th order statistic of the scores $r^{(j)}$, where $1-\alpha$ is the target nominal coverage level.

✓ **The Calibrated Interval:** The final, calibrated prediction interval for any new test point is constructed by inflating (or deflating) the raw interval bounds using λ_j:

Calibrated Band = $\hat{c}_j \pm \lambda_j \hat{h}_j$

✓ **Mathematical Guarantees:** Because the scores are exchangeable across the rows where the feature is missing, this split-conformal procedure provides a distribution-free, finite-sample guarantee. (1) The marginal coverage is mathematically guaranteed to be $\geq 1-\alpha$. (2) The realized coverage theoretically lies tightly within the bound $\left[1-\alpha, 1-\alpha+\frac{1}{n_j+1}\right]$ in expectation.

Experiments

❑ Comparison of Imputation Methods (1)

method	Bean	Magic	Shoppers	Letter	California	avg	rank
Mean	.994	.989	1.001	.996	.908	.978	5.6
Median	1.016	1.035	1.063	1.007	.961	1.016	7.0
KNN	.435	.897	1.003	.616	.939	.778	4.8
MICE	.503	.793	.917	.858	.742	.763	4.4
MissForest	.342	.703	.914	.474	.700	.627	2.4
MIWAE	.365	.687	.917	.619	.693	.656	2.8
GAIN	1.077	1.162	1.408	1.374	1.003	1.205	8.0
MissFlow	**.329**	**.668**	**.844**	**.449**	**.626**	**.583**	**1.0**

✓ MissFlow achieves **a perfect average rank of 1.0 across all five diverse benchmark datasets** under a baseline 30% MCAR setup.

✓ With an overall average RMSE of 0.583, MissFlow consistently outperforms top machine learning baselines like MissForest (0.627) and deep generative frameworks like MIWAE (0.656).

✓ The performance gap remains **robustly stable across different datasets and independently sampled missingness masks.**

Experiments

❑ Comparison of Imputation Methods (2)

method	generative	MI	UQ	calib. guar.	NFE	>MCAR
Mean / k-NN [16]	×	×	×	×	n/a	×
MICE [17]	∼	✓	∼	×	n/a	∼
MissForest [18]	×	×	×	×	n/a	×
GAIN [23]	✓	×	×	×	1	×
MIWAE [24]	✓	✓	∼	×	K	∼
TabCSDI [11]	✓	✓	∼	×	50–150	×
DiffPuter [12]	✓	✓	∼	×	50–150	×
MissFlow (ours)	✓	✓	✓	✓	≈20	✓

✓ While some can produce Multiple Imputations (MI) and Uncertainty Quantification (UQ), they completely lack mathematical guarantees for calibrated coverage.

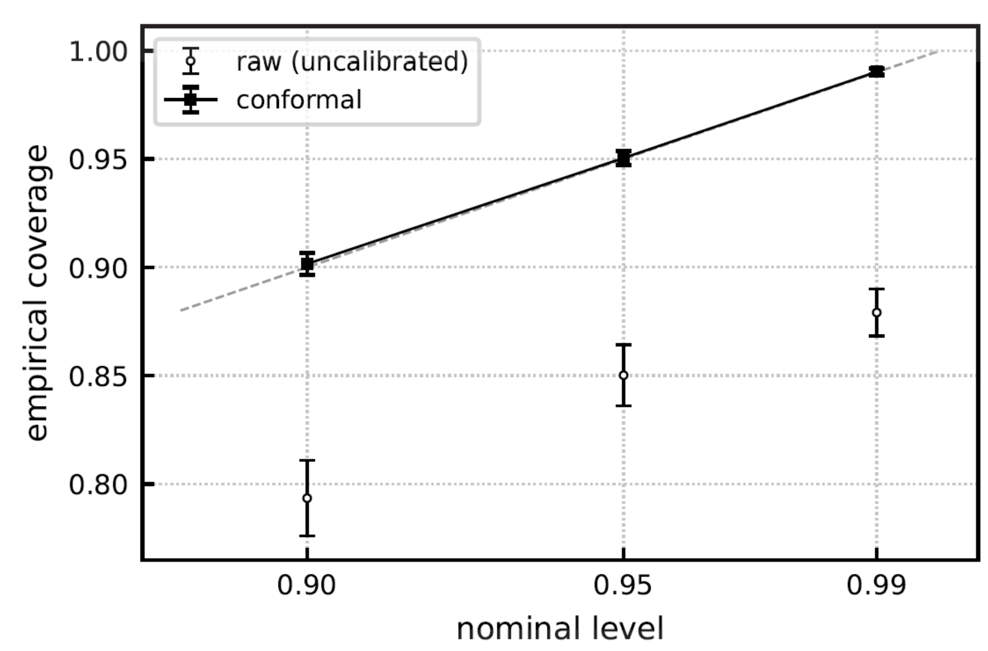
✓ These are strong generative models but are computationally heavy—requiring 50 to 150 NFEs per completion—and still fail to provide calibrated coverage guarantees.

✓ MissFlow is the only method in the comparison that delivers on all fronts. It is a fully generative model that offers **computationally cheap multiple imputations ($\approx$ 20 NFEs)**, is evaluated on complex missing data mechanisms beyond just MCAR, and uniquely provides a finite-sample calibrated coverage guarantee.

❑ Empirical Validation of Conformal Calibration

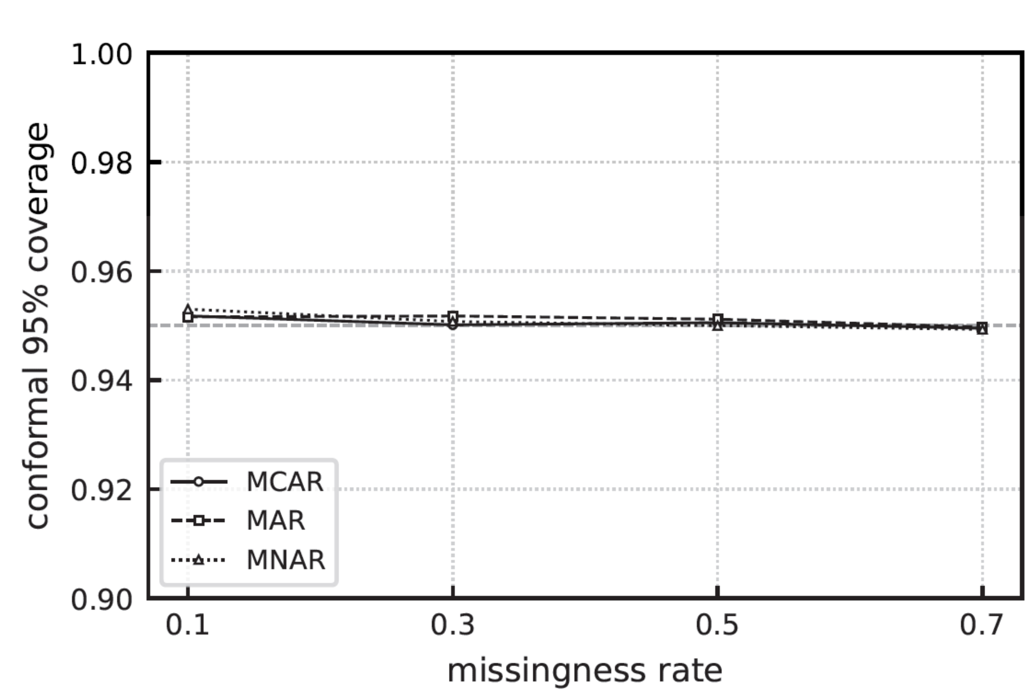
dataset	RMSE	raw cov	conformal cov	width
Bean	0.329	0.862±.011	0.949±.004	0.68
Magic	0.668	0.833±.001	0.951±.001	2.06
Shoppers	0.844	0.865±.009	0.950±.004	2.91
Letter	0.449	0.852±.003	0.951±.002	1.67
California	0.626	0.839±.005	0.951±.003	2.01

✓ Raw Multiple Imputation (MI) intervals from generative models naturally over-concentrate and under-cover the true missing values. Across all five datasets, the uncalibrated raw intervals achieve only ~83.3% to 86.5% empirical coverage when tasked with a 95% target.



✓ As illustrated, raw intervals fail uniformly across all target thresholds (under-covering at 90%, 95%, and 99%). The conformal calibration layer successfully pulls these points directly onto the **ideal diagonal line**.

❑ Robustness to Mechanism and Missingness Rate



✓ A common failure mode for traditional uncertainty estimation occurs when the missingness mechanism grows complex or data loss becomes severe.

✓ Above figure maps the calibrated coverage across an extensive sweep of missingness rates (10% to 70%) under three distinct settings: Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR).

✓ The calibrated 95% coverage line remains completely flat and anchored at 0.95 under all test conditions, empirically confirming **the distribution-free coverage guarantee in practice.**

Conclusion

✓ In this work, we present **MissFlow, a fast generative imputer for tabular data.** Using conditional flow matching, it performs multiple imputation via an ODE solver in just ~20 steps. To fix interval under-coverage, we apply **post-hoc split-conformal calibration**, providing a distribution-free nominal coverage guarantee alongside state-of-the-art accuracy.